# Eczema
# atopic dermatitis

Dr. Eman Gomaa

Consultant dermatology

# Eczema

# eczema

- 
  • "eczema " refers to a group of inflammatory dermatoses, of different etiologies, that share a distinctive clinical and histological features:
- 
  A Clinically, eczematous dermatoses are characterized by variable degrees of itching, soreness, dryness, erythema, scaling, excoriation, exudation, , lichenification .
- .
  A Histologically, eczema is characterized by a range of epidermal changes including spongiosis with varying degrees of acanthosis, hyperkeratosis, and lymphohistiocytic infiltrate in the dermis.

eczema

Endogenouse eczema

Exogenouse eczema

**Endogenous eczemas**

1. Atopic dermatitis
2. Seborrhoeic dermatitis
3. Asteatotic eczema
4. Venous eczema
5. Pityriasis alba
6. Discoid eczema
7. Eyelid eczema
8. Hand eczema
9. Juvenile plantar dermatosis

**Exogenouse eczema**

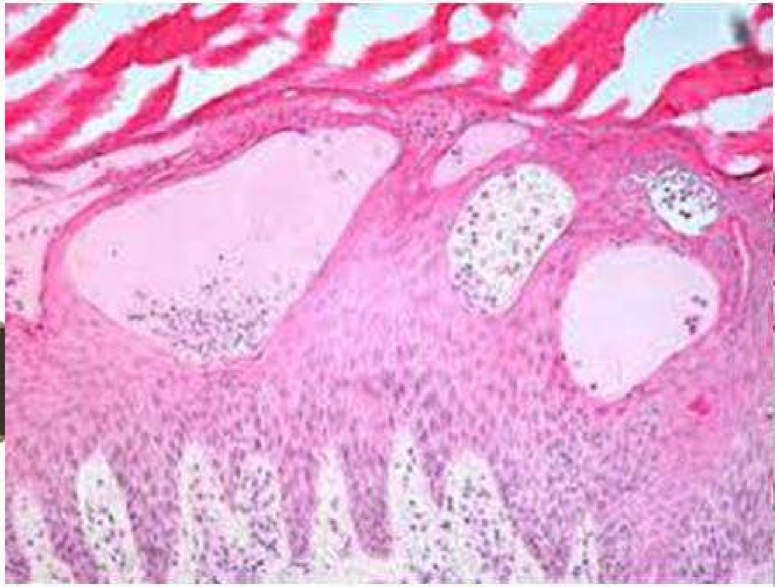
Contact dermatitis
- Irritant
- Allergic

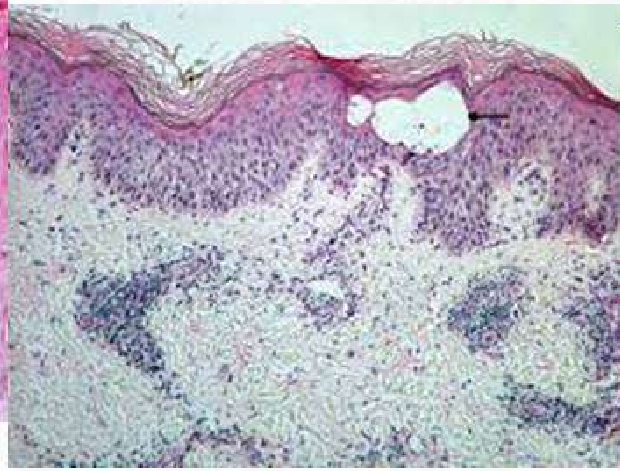
2. Photoallergic contact dermatitis
3. Infective eczema
4. Dermatophytide
5. Eczematous polymorphic light eruption

## Clinical classification (acute, subacute and chronic eczema)

| Acute | Subacute | Chronic |
|---|---|---|
| ***Clinically:***<br>Erythema; papules, vesicles and bullae; oedema, exudation (oozing) | Features are midway between those of acute and chronic types | Dryness and ***lichenification*** (thickened hyperpigmented skin with accentuated skin markings) |
| ***Histopathologically:***<br>***Spongiotic***<br>• Normal basket-weave stratum corneum<br>• **Spongiosis**<br>• **Exocytosis**<br>• Papillary derma edema | ***Psoriasiform-spongiotic***<br>• **Parakeratosis**<br>• Little spongiosis<br>• **Acanthosis**<br>• Little to no papillary dermal edema | ***Psoriasiform***<br>• **Compact hyperkeratosis**<br>• More prominent **acanthosis** (lichenification) |
| • Superficial perivascular inflammatory infiltrate of lymphocytes | | |


Acute eczema
• Acute: pruritus, erythema, vesiculation

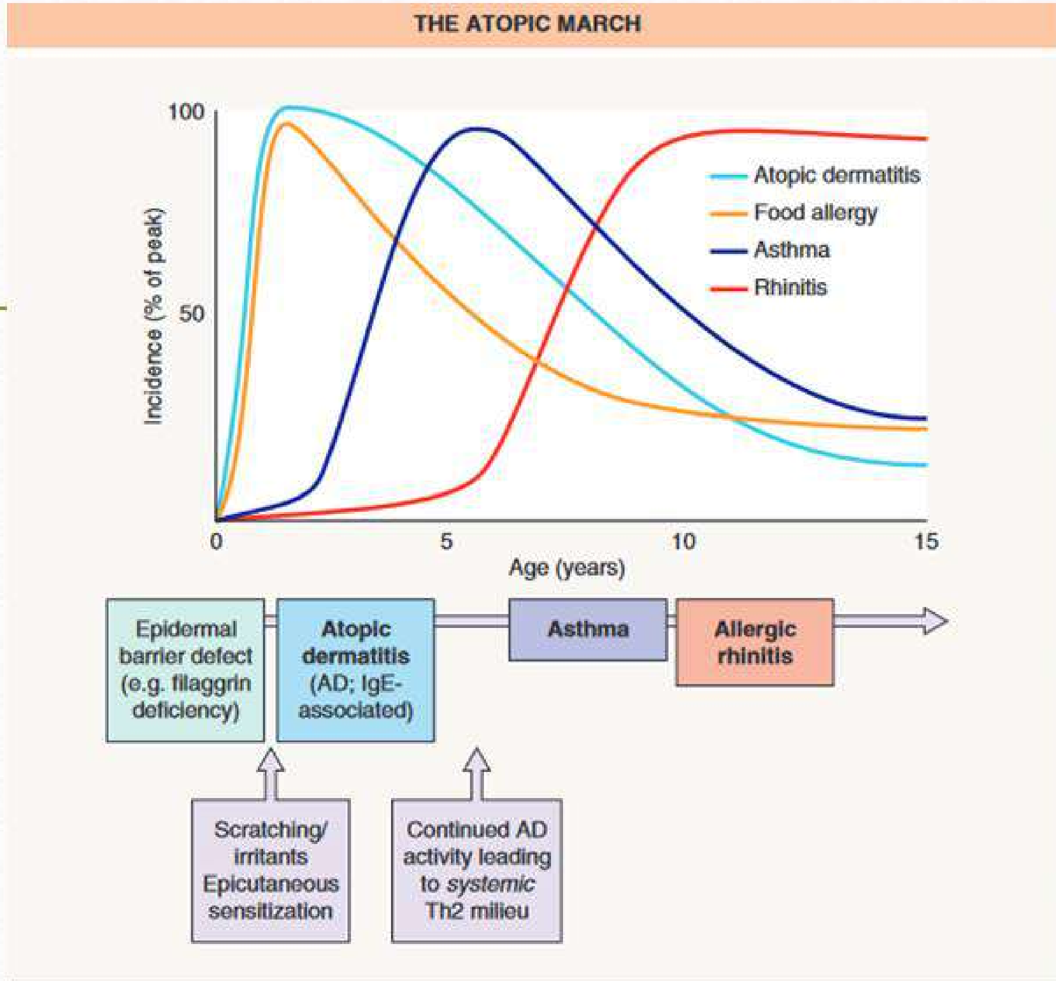




# Atopic dermatitis


**Fig. 12.7 Flexural atopic dermatitis in a child.** The popliteal fossa is a typical location. Note the excoriations. *Courtesy, Julie V Schaffer, MD.*

- Atopy :inherited tendency to develop allergy towards common environmental allergens.
- Triad:
  - atopic dermatitis,
  - Allergic rhino-conjunctivitis
  - bronchial asthma.


THE ATOPIC MARCH
100
50
Incidence (% of peak)
0
5
10
15
Age (years)
Atopic dermatitis
Food allergy
Asthma
Rhinitis
Epidermal barrier defect (e.g. filaggrin deficiency)
Atopic dermatitis (AD; IgE-associated)
Asthma
Allergic rhinitis
Scratching/ irritants Epicutaneous sensitization
Continued AD activity leading to systemic Th2 milieu

- Atopic dermatitis : is a common, chronic, or chronically relapsing, inflammatory skin condition that typically begins during infancy or early childhood.
- Clinically, it is characterized by itchy papules which become excoriated and lichenified typically in a flexural distribution.

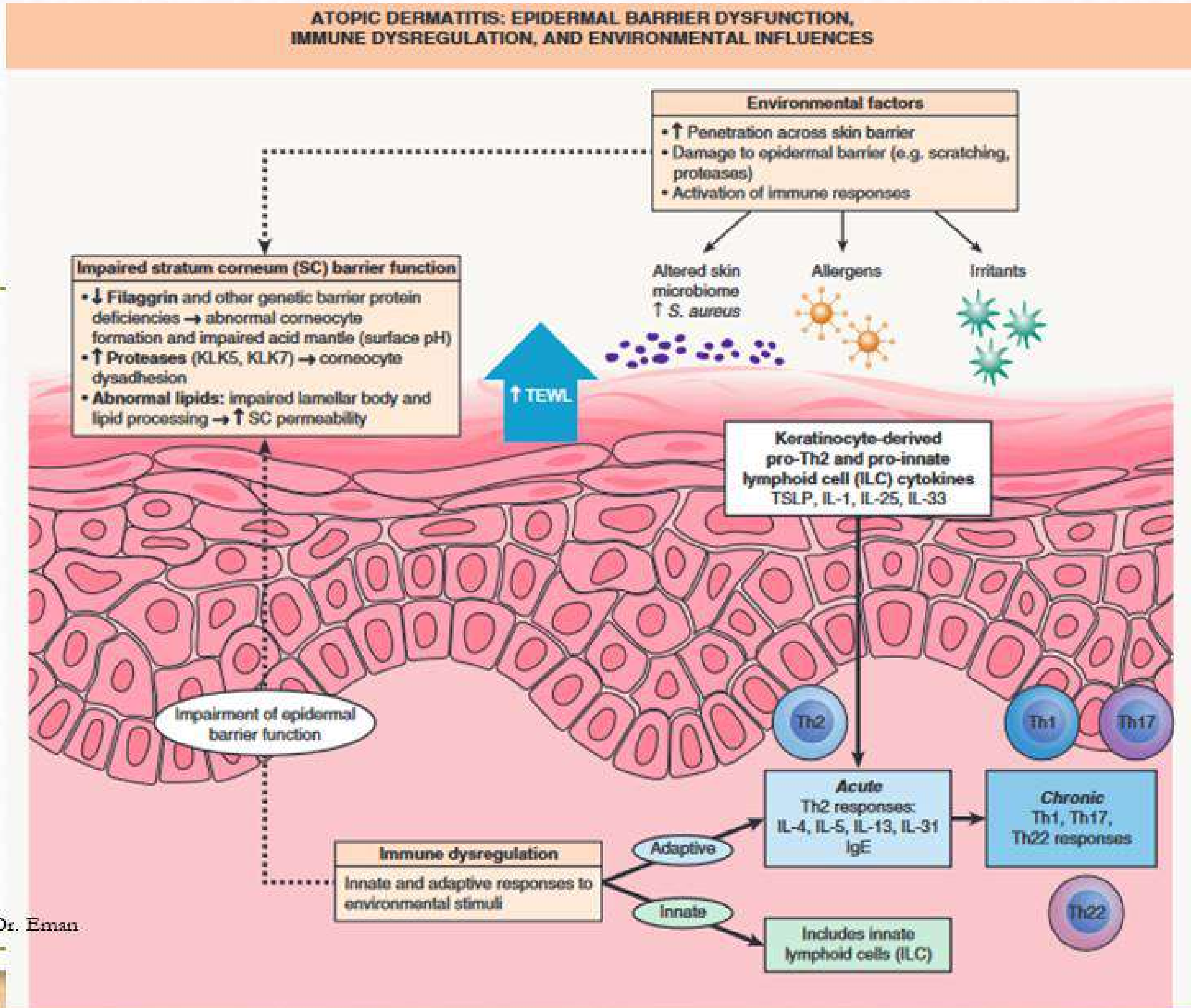
# Aetiopathogensis

Enviromental triggers

In Genaticly susptable pt

Immune dysregulation

Skin barrier


ATOPIC DERMATITIS: EPIDERMAL BARRIER DYSFUNCTION, IMMUNE DYSREGULATION, AND ENVIRONMENTAL INFLUENCES
Environmental factors
• ↑ Penetration across skin barrier
• Damage to epidermal barrier (e.g. scratching, proteases)
• Activation of immune responses
Altered skin microbiome ↑ S. aureus
Allergens
Irritants
Impaired stratum corneum (SC) barrier function
• ↓ Filaggrin and other genetic barrier protein deficiencies → abnormal corneocyte formation and impaired acid mantle (surface pH)
• ↑ Proteases (KLK5, KLK7) → corneocyte dysadhesion
• Abnormal lipids: impaired lamellar body and lipid processing → ↑ SC permeability
↑ TEWL
Keratinocyte-derived pro-Th2 and pro-innate lymphoid cell (ILC) cytokines TSLP, IL-1, IL-25, IL-33
Impairment of epidermal barrier function
Th2
Th1
Th17
Acute Th2 responses: IL-4, IL-5, IL-13, IL-31 IgE
Chronic Th1, Th17, Th22 responses
Immune dysregulation
Innate and adaptive responses to environmental stimuli
Adaptive
Innate
Includes innate lymphoid cells (ILC)
Th22

# Genatic predesposition

- higher among monozygotic twins (77%) compared to dizygotic twins (15%).
- 

  • Positive family history (atopic disorders are more frequently transmitted by mothers than by fathers).
- 

- Genes implicated in the pathogenesis o f AD :
  1. Genes encoding epidermal proteins (specific to the skin) e.g.
- filaggrin gene
- SPINK5 “serine peptidase inhibitor Kazal type 5”) —> impaired epidermal barrier

# Skin barrier

- Deficiency of filaggrin —>impaired keratin filaments aggregation and water-binding capacity of stratum comeum.
- 
  • Changes in the balance between proteolytic enzymes (e.g. peptidase) and their anti-protease counterparts (e.g. SPINK5) —► increase in protease activity —>accelerates degradation of corneodesmosomes and lipid-processing enzymes —>barrier breakdown and failure to generate ceramides.
- 
  • Disturbed maturation of lamellar bodies —>alterations in epidermal lipid metabolism and composition.

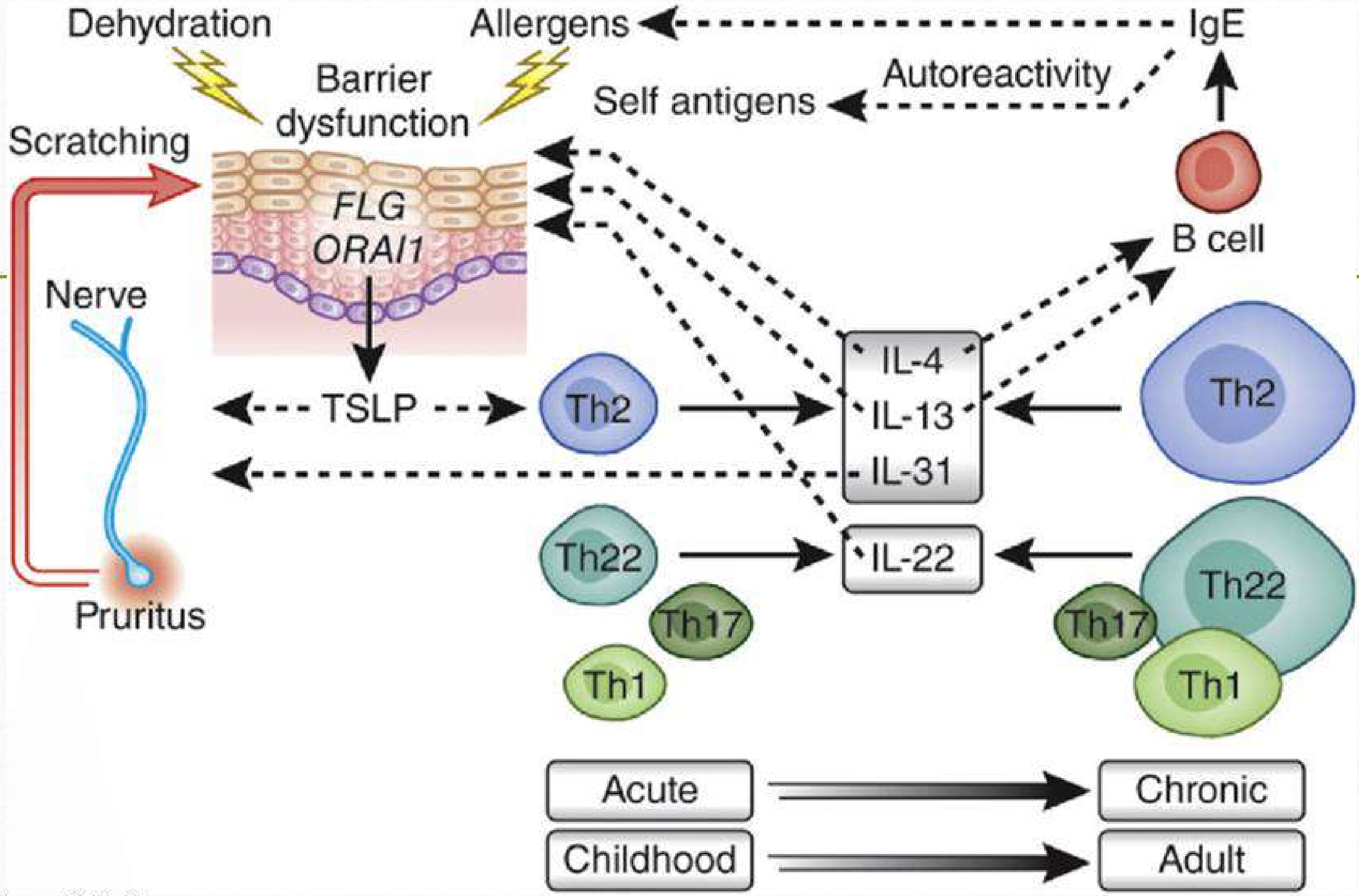
- Epidermal barrier dysfunction resulting in
- 
  - Dry scaly skin (xerosis)
- 
  - Increased penetration of environmental irritants and allergens

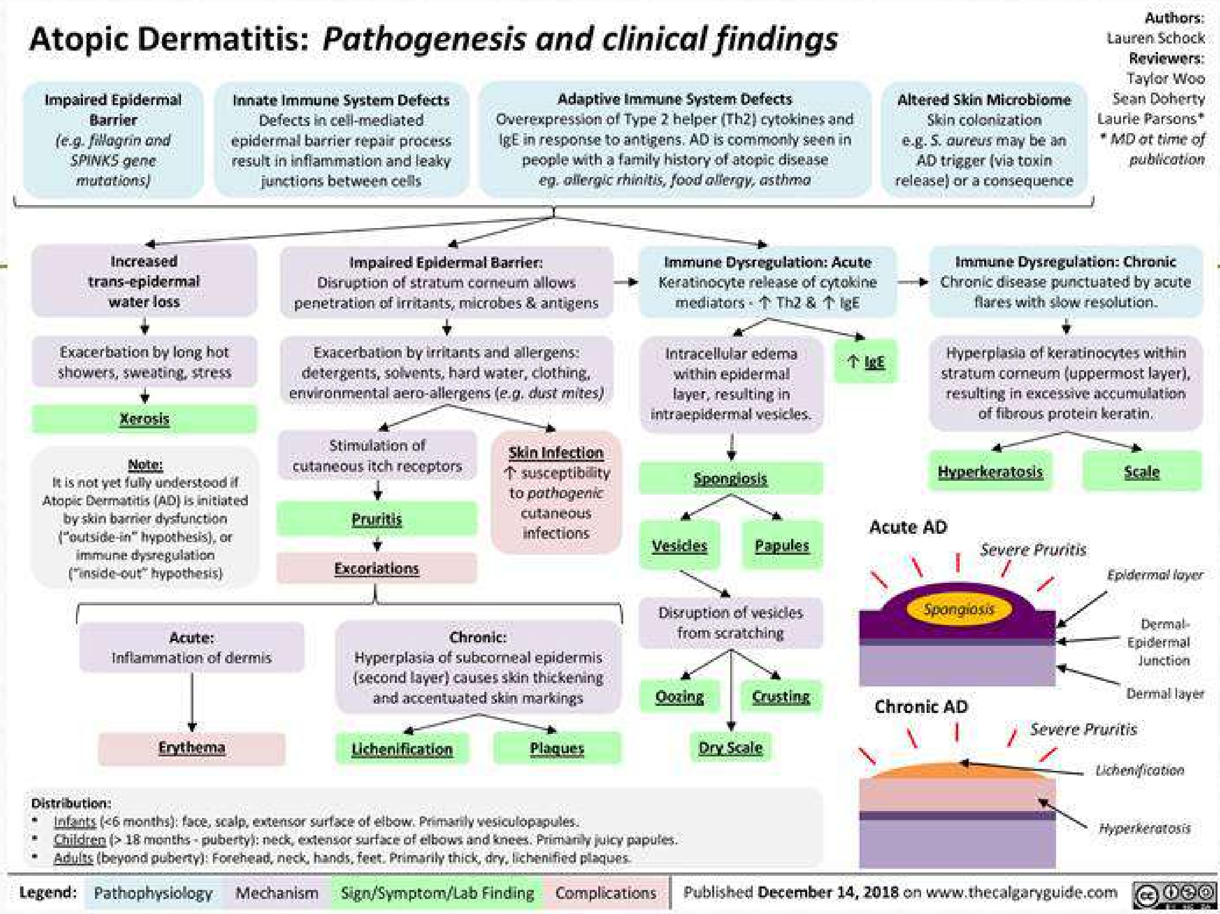
- Environm ental factors:
-
  o Infectious agents: S. aureus , MC , HS
-
  o Other triggers: Allergens (e.g. pollen, house dust mites, animal dander),
-
  Sweating, Harsh soaps, Wool or other rough fabrics, Cigarette smoke, Emotional stress and
- Food (eggs, milk, peanuts, soy and wheat).

# Immune dysregulation

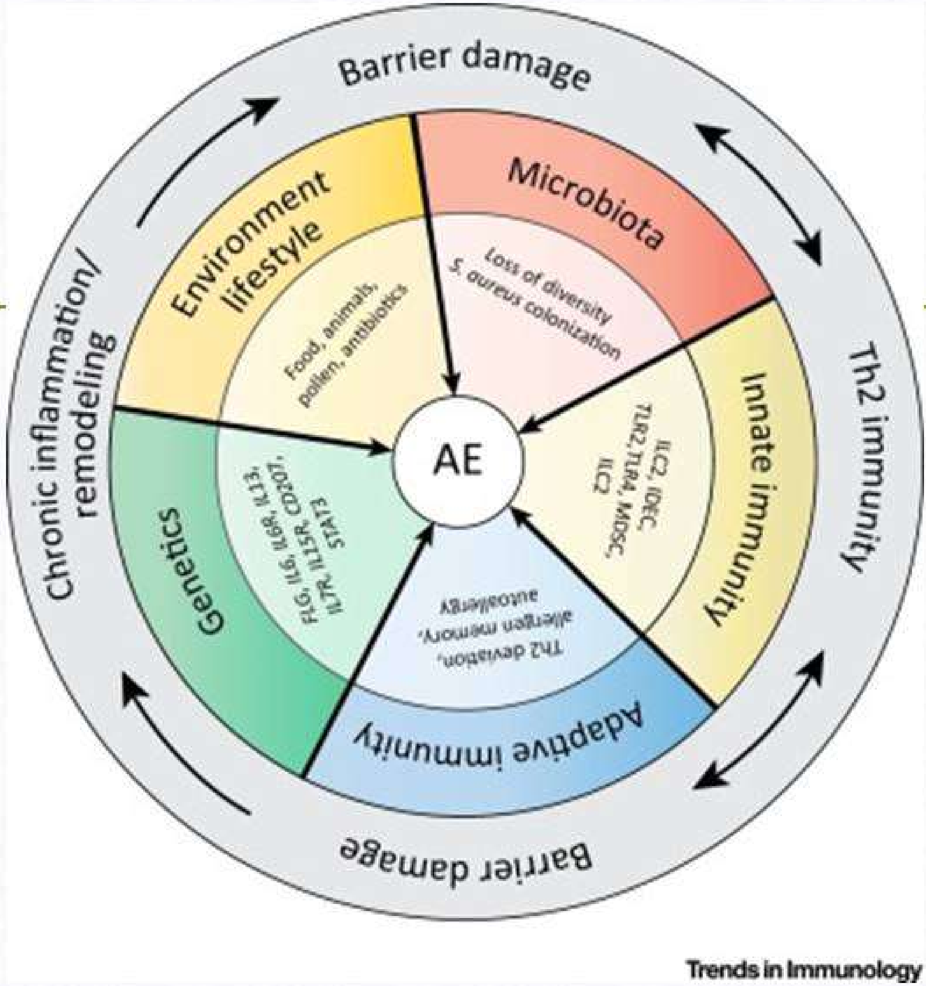
o Innate immunity: decreased production of antimicrobial peptides,

o Acquired immunity:
o Th2 cytokines (e.g. IL-4, IL-5, IL-13) predominate in acute AD lesions,
o Thl cytokines (e.g. IFN-y, IL-12) predominate in chronic AD lesions.


Dehydration
Allergens
IgE
Autoreactivity
Barrier dysfunction
Self antigens
Scratching
FLG
ORAI1
B cell
Nerve
IL-4
TSLP
Th2
IL-13
Th2
IL-31
Th22
IL-22
Th22
Pruritus
Th17
Th17
Th1
Th1
Acute
Chronic
Childhood
Adult



Atopic Dermatitis: Pathogenesis and clinical findings
Authors:
Lauren Schock
Reviewers:
Taylor Woo
Sean Doherty
Laurie Parsons*
* MD at time of publication
Impaired Epidermal Barrier
(e.g. fillagrin and SPINK5 gene mutations)
Innate Immune System Defects
Defects in cell-mediated epidermal barrier repair process result in inflammation and leaky junctions between cells
Adaptive Immune System Defects
Overexpression of Type 2 helper (Th2) cytokines and IgE in response to antigens. AD is commonly seen in people with a family history of atopic disease
eg. allergic rhinitis, food allergy, asthma
Altered Skin Microbiome
Skin colonization
e.g. S. aureus may be an AD trigger (via toxin release) or a consequence
Increased trans-epidermal water loss
Exacerbation by long hot showers, sweating, stress
Xerosis
Note:
It is not yet fully understood if Atopic Dermatitis (AD) is initiated by skin barrier dysfunction ("outside-in" hypothesis), or immune dysregulation ("inside-out" hypothesis)
Impaired Epidermal Barrier:
Disruption of stratum corneum allows penetration of irritants, microbes & antigens
Exacerbation by irritants and allergens: detergents, solvents, hard water, clothing, environmental aero-allergens (e.g. dust mites)
Stimulation of cutaneous itch receptors
Pruritis
Excoriations
Skin Infection
↑ susceptibility to pathogenic cutaneous infections
Acute:
Inflammation of dermis
Erythema
Chronic:
Hyperplasia of subcorneal epidermis (second layer) causes skin thickening and accentuated skin markings
Lichenification
Plaques
Immune Dysregulation: Acute
Keratinocyte release of cytokine mediators - ↑ Th2 & ↑ IgE
Intracellular edema within epidermal layer, resulting in intraepidermal vesicles.
↑ IgE
Spongiosis
Vesicles
Papules
Disruption of vesicles from scratching
Oozing
Crusting
Dry Scale
Immune Dysregulation: Chronic
Chronic disease punctuated by acute flares with slow resolution.
Hyperplasia of keratinocytes within stratum corneum (uppermost layer), resulting in excessive accumulation of fibrous protein keratin.
Hyperkeratosis
Scale
Acute AD
Severe Pruritis
Spongiosis
Epidermal layer
Dermal-Epidermal Junction
Dermal layer
Chronic AD
Severe Pruritis
Lichenification
Hyperkeratosis
Distribution:
• Infants (<6 months): face, scalp, extensor surface of elbow. Primarily vesiculopapules.
• Children (> 18 months - puberty): neck, extensor surface of elbows and knees. Primarily juicy papules.
• Adults (beyond puberty): Forehead, neck, hands, feet. Primarily thick, dry, lichenified plaques.
Legend: Pathophysiology Mechanism Sign/Symptom/Lab Finding Complications
Published December 14, 2018 on www.thecalgaryguide.com


Barrier damage
Microbiota
Loss of diversity
S. aureus colonization
Th2 immunity
Innate immunity
ILC2, IDEC,
TLR2,TLR4, MDSC,
ILC2
Adaptive immunity
Th2 deviation,
allergen memory,
autoallergy
Barrier damage
Genetics
FLG, IL6, IL6R, IL13,
IL7R, IL15R, CD207,
STAT3
Chronic inflammation/
remodeling
Environment
lifestyle
Food, animals,
pollen, antibiotics
AE
Trends in Immunology

- Symptoms:
  o Persistent, severe pruritus (AD is known as the "itch that rashes").

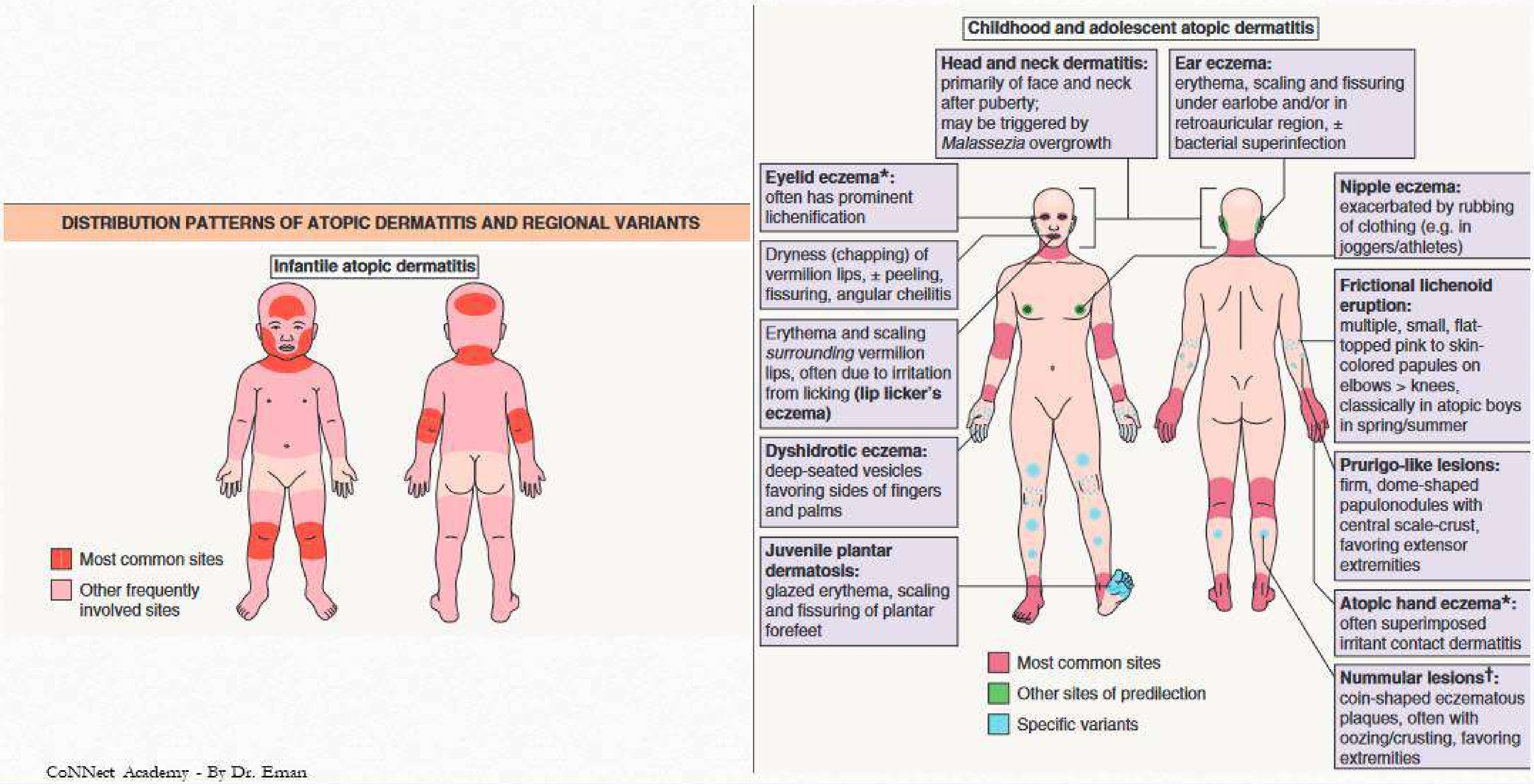
## • Skin lesions:

- o **Acute lesions**: (infantile stage) - edematous, erythematous papules and plaques with vesiculation, oozing and serous crusting
- o **Subacute lesions**: erythema, scaling and variable crusting,
- o **Chronic lesions**: (adolescent/adult stage) - thickened plaques with lichenification, scaling prurigo nodularis - like lesions,
- o Dry skin (Xerosis)


A
B
C

# Stages of AD

- o Infantile stage (>2 months - < 2 years):
  Edematous papules and papulo-vesicles on the cheeks —> large plaques with oozing and crusting,
- o Childhood stage (age 2 - 1 2 years):
  The lesions are less exudative and tend to become lichenified.
  The classic sites of predilection are the antecubital and popliteal fossae (flexural eczema)
- ,
  o Adult/adolescent AD (age >12 years):
  Subacute to chronic lichenified lesions. The clinical picture is similar to that of later childhood stage with involvement of the flexural folds


DISTRIBUTION PATTERNS OF ATOPIC DERMATITIS AND REGIONAL VARIANTS
Infantile atopic dermatitis
Most common sites
Other frequently involved sites
Childhood and adolescent atopic dermatitis
Head and neck dermatitis: primarily of face and neck after puberty; may be triggered by Malassezia overgrowth
Ear eczema: erythema, scaling and fissuring under earlobe and/or in retroauricular region, ± bacterial superinfection
Eyelid eczema*: often has prominent lichenification
Dryness (chapping) of vermilion lips, ± peeling, fissuring, angular cheilitis
Erythema and scaling surrounding vermilion lips, often due to irritation from licking (lip licker's eczema)
Dyshidrotic eczema: deep-seated vesicles favoring sides of fingers and palms
Juvenile plantar dermatosis: glazed erythema, scaling and fissuring of plantar forefeet
Nipple eczema: exacerbated by rubbing of clothing (e.g. in joggers/athletes)
Frictional lichenoid eruption: multiple, small, flat-topped pink to skin-colored papules on elbows > knees, classically in atopic boys in spring/summer
Prurigo-like lesions: firm, dome-shaped papulonodules with central scale-crust, favoring extensor extremities
Atopic hand eczema*: often superimposed irritant contact dermatitis
Nummular lesions†: coin-shaped eczematous plaques, often with oozing/crusting, favoring extremities
Most common sites
Other sites of predilection
Specific variants

# COMPLICATION OF AD

- Infection :staph ,eczemz herpiticum
- Erythroderma
- Growth delay
- Psychological stress
- Eye
  - Keratoconjunctivitis , keratoconus
  - Cataract , retinal detatchment
  - Dennie morgan fold
  - darkning

# Diagnostic criteria
# hanifin , rajka 1980

| Major criteria (3 or more required) | Minor criteria (3 or more required) |
| --- | --- |
| Pruritus<br>Typical morphology and distribution<br>• Flexural lichenification or linearity in adults<br>• Facial and extensor involvement in infants and children<br>Chronic or chronically-relapsing dermatitis<br>Personal or family history of atopy (asthma, allergic rhinitis, atopic dermatitis) | Xerosis<br>Ichthyosis, palmar hyperlinearity, or keratosis pilaris<br>Immediate (type 1) skin-test reactivity<br>Raised serum IgE<br>Early age of onset<br>Tendency toward cutaneous infections (especially S. *aureus* and herpes simplex) or impaired cell-mediated immunity<br>Tendency toward non-specific hand or foot dermatitis<br>Nipple eczema<br>Cheilitis<br>Recurrent conjunctivitis<br>Dennie-Morgan infraorbital fold<br>Keratoconus<br>Anterior subcapsular cataracts<br>Orbital darkening<br>Facial pallor or facial erythema<br>Pityriasis alba<br>Anterior neck folds<br>Itch when sweating<br>Intolerance to wool and lipid solvents<br>Perifollicular accentuation<br>Food intolerance<br>Course influenced by environmental or emotional factors<br>White dermographism or delayed blanch |

**DIAGNOSTIC FEATURES AND TRIGGERS OF ATOPIC DERMATITIS (AD)**

**Essential features: must be present and are sufficient for diagnosis**

- Pruritus
  - Rubbing or scratching can initiate or exacerbate flares ("the itch that rashes")
  - Often worse in the evening and triggered by exogenous factors (e.g. sweating, rough clothing)
- Typical eczematous morphology and age-specific distribution patterns (see Fig. 12.3)
  - Face, neck, and extensor extremities in infants and young children
  - Flexural lesions at any age
  - Sparing of groin and axillae
- Chronic or relapsing course

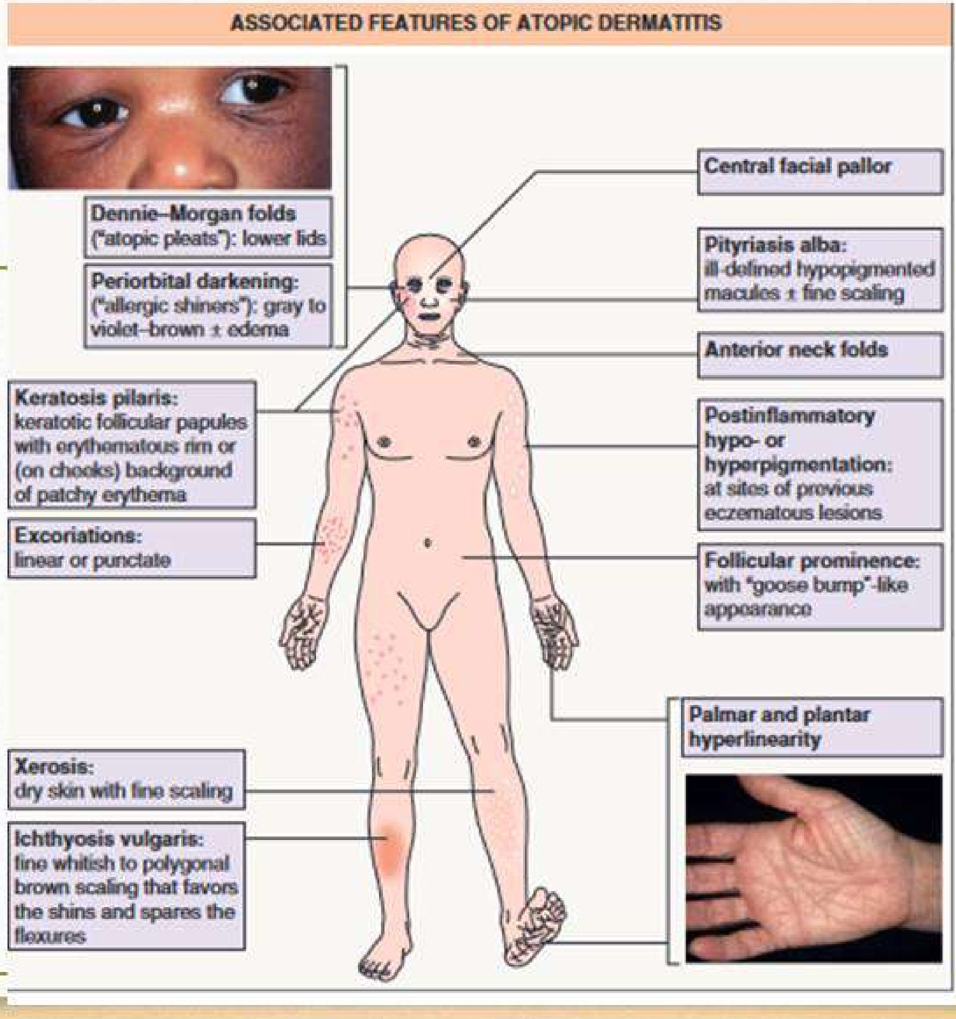
**Important features: seen in most cases, supportive of diagnosis**

- Onset during infancy or early childhood
- Personal and/or family history of atopy (IgE reactivity)
- Xerosis
  - Dry skin with fine scale in areas *without* clinically apparent inflammation; often leads to pruritus

**Associated features: suggestive of the diagnosis, but less specific (see Figs 12.3 & 12.14)**

- Other filaggrin deficiency-associated conditions: keratosis pilaris, hyperlinear palms, ichthyosis vulgaris
- Follicular prominence, lichenification, prurigo lesions
- Ocular findings: recurrent conjunctivitis, anterior subcapsular cataract; periorbital changes: pleats, darkening
- Other regional findings, e.g. perioral or periauricular dermatitis, pityriasis alba
- Atypical vascular responses, e.g. midfacial pallor, white dermographism*, delayed blanch

**Triggers**


ASSOCIATED FEATURES OF ATOPIC DERMATITIS
Dennie–Morgan folds ("atopic pleats"): lower lids
Periorbital darkening: ("allergic shiners"): gray to violet–brown ± edema
Keratosis pilaris: keratotic follicular papules with erythematous rim or (on cheeks) background of patchy erythema
Excoriations: linear or punctate
Xerosis: dry skin with fine scaling
Ichthyosis vulgaris: fine whitish to polygonal brown scaling that favors the shins and spares the flexures
Central facial pallor
Pityriasis alba: ill-defined hypopigmented macules ± fine scaling
Anterior neck folds
Postinflammatory hypo- or hyperpigmentation: at sites of previous eczematous lesions
Follicular prominence: with "goose bump"-like appearance
Palmar and plantar hyperlinearity

# DD

| DIFFERENTIAL DIAGNOSIS OF ATOPIC DERMATITIS | | |
|---|---|---|
| **Chronic dermatoses** | | |
| **C > A** | Seborrheic dermatitis | Common |
| **B** | Contact dermatitis – allergic* or irritant | Common |
| **B** | Psoriasis | Common |
| **A > C** | Nummular dermatitis | Uncommon (although nummular lesions can be seen in AD, nummular eczema outside this setting is uncommon) |
| **A** | Asteatotic eczema | Common |
| **A > C** | Lichen simplex chronicus (secondary to pruritus of variable etiology) | Common |
| **Infections and infestations** | | |
| **B** | Scabies | Common |
| **B** | Dermatophytosis* | Common |
| **B** | HIV-associated dermatoses | Variable |
| **C** | HTLV-1-associated "infective dermatitis" | Variable |
| **C** | Chronic mucocutaneous candidiasis | Rare |
| **C** | Neonatal mucocutaneous candidiasis | Variable |
| **B** | Impetigo | Variable (nummular lesions of AD can mimic impetigo) |
| **C** | Congenital syphilis | Variable |
| **Primary immunodeficiencies** | | |
| **C** | Wiskott–Aldrich syndrome | Rare |
| **C** | Hyperimmunoglobulin E syndromes | Rare |
| **C** | IPEX (*i*mmune dysregulation, *p*olyendocrinopathy, *e*nteropathy, *X*-linked) syndrome and IPEX-like conditions | Rare |
| **C** | Omenn syndrome | Rare |
| **C** | DiGeorge syndrome (congenital thymic aplasia) | Rare |
| **C** | Immunoglobulin deficiencies: X-linked hypogammaglobulinemia, common variable immunodeficiency, IgA deficiency | Variable |
| **C** | Ataxia telangiectasia | Rare |

| | | |
|---|---|---|
| **Malignancies** | | |
| **A > C** | Mycosis fungoides and Sézary syndrome | Uncommon |
| **C** | Langerhans cell histiocytosis | Rare |
| **Metabolic and genetic disorders** | | |
| **C** | Netherton syndrome, generalized inflammatory peeling skin syndrome | Rare |
| **C** | Ectodermal dysplasias | Rare |
| **C** | *S*evere *d*ermatitis–*a*llergies–*m*etabolic wasting (SAM) syndrome | Rare |
| **B** | Keratosis pilaris | Common |
| **C** | Hartnup disease | Rare |
| **C** | Phenylketonuria | Rare |
| **C** | Prolidase deficiency | Rare |
| **B** | Acrodermatitis enteropathica, glucagonoma syndrome, pellagra, riboflavin deficiency | Rare |
| **C > A** | Other causes of "nutritional dermatitis", e.g. biotin deficiency, essential fatty acid deficiency, organic acidurias, cystic fibrosis | Rare |
| **Autoimmune disorders** | | |
| **A > C** | Dermatitis herpetiformis | Uncommon |
| **A > C** | Pemphigus foliaceus | Uncommon |
| **B** | Dermatomyositis | Uncommon (but often misdiagnosed as AD in children) |
| **A > C** | Lupus erythematosus | Uncommon |
| **Other** | | |
| **B** | Drug eruptions | Uncommon (although drug eruptions are common, those resembling atopic dermatitis are uncommon) |
| **A > C** | Photoallergic contact dermatitis | Uncommon |
| **A** | Chronic actinic dermatitis | Uncommon |
| **B** | Graft-versus-host disease | Uncommon |

- Aims of treatment:
  Because AD is a chronic relapsing disease, the aims of treatment are (1) targeting
  acute flares with short-term treatment regimens (reactive management) and (2)
  long-term maintenance therapy (to avoid relapses) i.e.
- control rather than cure.

## TREATMENT

| PREVENTION | EDUCATION |
|---|---|

| AVOID TRIGGERING FACTOR | SKIN CARE |
|---|---|

## ANTI INLAMATORY THERAPY

# 1. PREVENTION

- Protection against the development of AD in high-risk infants
  - . Exclusive breastfeeding during the first 4 - 6 months of life
  - A low-allergen maternal diet during breastfeeding
  - Administration of probiotics (e.g. lactobacilli) or prebiotics (non-digestible oligosaccharides that promote the growth of desirable bacteria) to pregnant mothers and infants

# EDUCATION

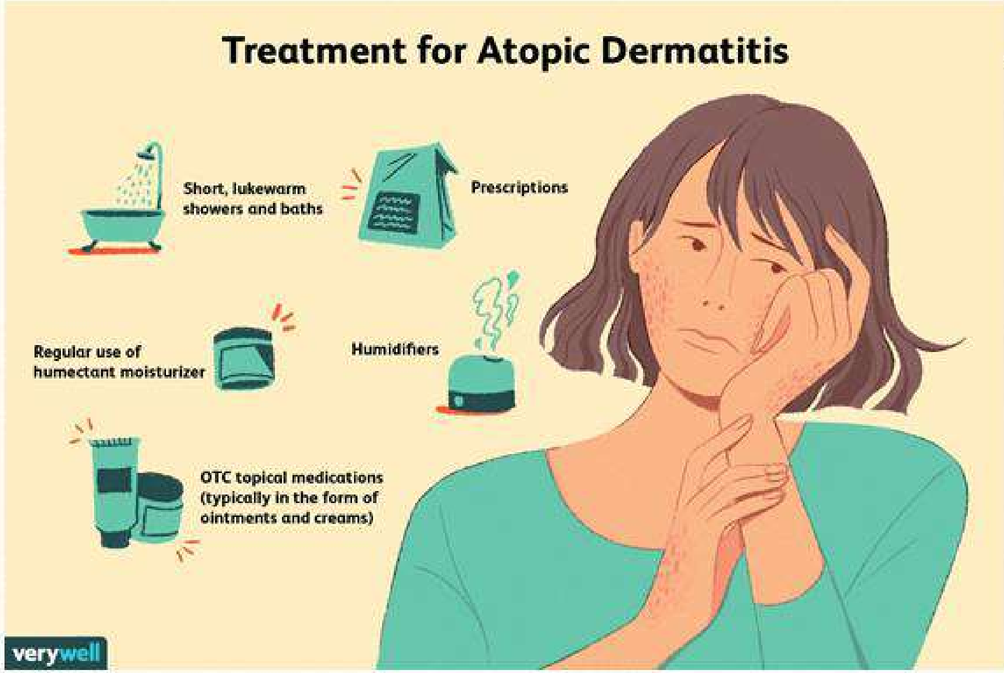
- Education of patients and their parents about
  - triggers / natural history of the disease
  - skin care
  - the treatment strategy: “control” rather than “cure”
  - reassurance about corticosteroid use (parents are anxious about steroids side effects.

# Identificatin ,AVOID TRIGGERING FACTORS

- careful history
- - Allergy testing (when indicated)
  - o Fluorescence enzyme immunoassays :to quantify specific IgE antibodies against suspected allergens,
  - Skin prick testing to assess for immediate hypersensitivity,
  - o The atopy patch test to detect delayed hypersensitivity.

- DECREASE HOUSE DUST MITE
- Decrease food hyper sensitivity :
  - Avoidance of food allergens
- Decrease staph aureus colonization
  - Cleansers and emollients containing antiseptics .
    - 0.005% sodium hypochlorite baths (0.5 cup of household bleach [6% sodium hypochlorite] added to a full 40-gallon bathtub) twice weekly

# Skin care



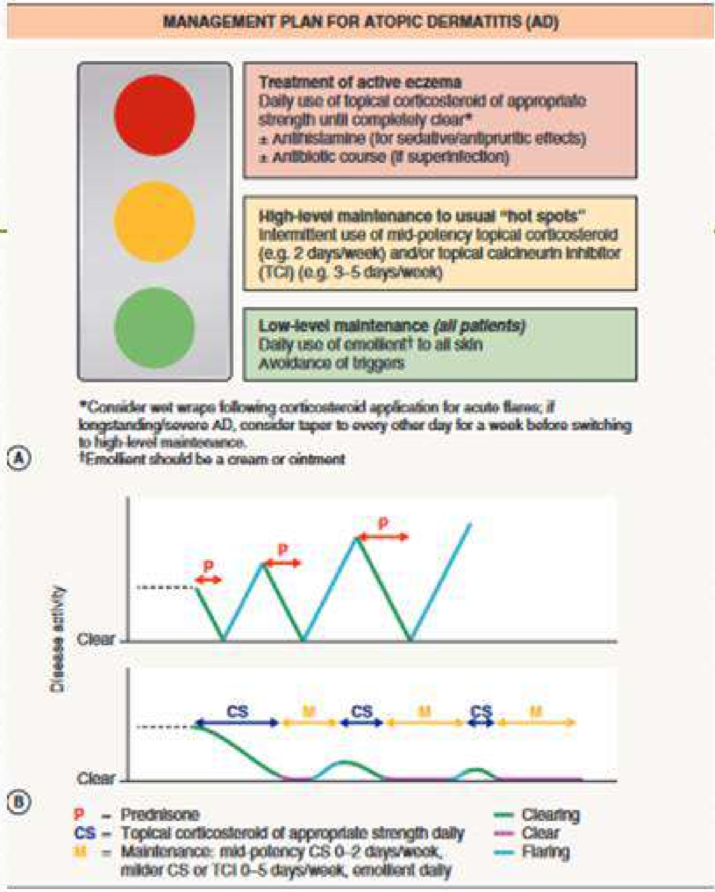

- emollients:

  - Emollients should be applied twice daily to the entire cutaneous surface.

- Bathing behavior
  - Tap water
  - Mild non alkaline cleanser
  - Emolient after path

# Antinflammatory therapy ( control flares)

- Oral antihistamines (antipruritic / sedative effects)
- • Topical therapy (localized lesions): corticosteroids or calcineurin inhibitors
- • Phototherapy (generalized lesions): BB-UVB, NB-UVB, UVA1 or PUVA.
- • Systemic corticosteroids (short term for severe acute flares)
- • Steroid-sparing agents (severe recalcitrant lesions) e.g. cyclosporine, azathioprine, mycophenolate mofetil, methotrexate, interferon-y, IVIG,


MANAGEMENT PLAN FOR ATOPIC DERMATITIS (AD)
Treatment of active eczema
Daily use of topical corticosteroid of appropriate strength until completely clear*
± Antihistamine (for sedative/antipruritic effects)
± Antibiotic course (if superinfection)
High-level maintenance to usual "hot spots"
Intermittent use of mid-potency topical corticosteroid (e.g. 2 days/week) and/or topical calcineurin inhibitor (TCI) (e.g. 3-5 days/week)
Low-level maintenance (all patients)
Daily use of emollient† to all skin
Avoidance of triggers
*Consider wet wraps following corticosteroid application for acute flares; if longstanding/severe AD, consider taper to every other day for a week before switching to high-level maintenance.
†Emollient should be a cream or ointment
A
Disease activity
Clear
P
CS
M
B
P = Prednisone
CS = Topical corticosteroid of appropriate strength daily
M = Maintenance: mid-potency CS 0-2 days/week, milder CS or TCI 0-5 days/week, emollient daily
Clearing
Clear
Flaring

| THERAPEUTIC LADDER FOR ATOPIC DERMATITIS | Evidence |
|---|---|
| Moisturizers | 1 |
| Topical corticosteroids | 1 |
| Topical calcineurin inhibitors | 1 |
| Topical crisaborole | 1 |
| Topical tofacitinib (not currently FDA-approved) | 1 |
| Narrowband UVB, UVA–UVB, UVA1 | 1 |
| Dupilumab | 1 |
| Cyclosporine (short/intermediate term) | 1 |
| Azathioprine | 1 |
| Mycophenolate mofetil/enteric-coated mycophenolate sodium | 1*/2 |
| Methotrexate | 1*/2 |
| Systemic corticosteroids (short term for severe acute flares; "rebound" exacerbations often occur upon discontinuation) | 2 |
| Omalizumab | 2† |
| Nemolizumab (anti-IL-31 receptor A; not currently FDA approved) | 1¶ |
| Tofacitinib | 2 |
| Rituximab | 2 |
| Interferon-γ | ** |
| IVIg | 2† |
| Other – crude coal tar, hydroxychloroquine, extracorporeal photochemotherapy | 2–3 |
| **Adjunctive therapies** | |
| Wet wraps, open wet dressings, or soaks combined with topical corticosteroids for acute flares<br>Dilute sodium hypochlorite (bleach) baths<br>Treatment of associated bacterial, viral, or fungal infections<br>Oral antihistamines for antipruritic‡ and sedative effects | |


Thank you !